

Sanskriti Mishra

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Professional Summary

Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning. Experienced in building AI-powered applications spanning computer vision, natural language processing, and large language model (LLM) integration, with hands-on expertise in Python, FastAPI, and applied machine learning pipelines. Strong foundation in data structures, algorithms, and object-oriented programming, with a track record of shipping end-to-end, production-oriented projects.

Technical Skills

Programming Languages: Python, Java

Machine Learning & AI: Machine Learning, Computer Vision, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Model Training, Model Evaluation

Frameworks & Libraries: FastAPI, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Databases: SQLite, MySQL

Developer Tools: Git, GitHub, Jupyter Notebook, Google Colab, Claude Code, GitHub Copilot, Vercel, MS Office

Projects

Breast Cancer Diagnosis using Machine Learning

[GitHub](#)

Python, Scikit-learn, Pandas, SVM, Random Forest

- Developed a binary classification model using the Wisconsin Breast Cancer dataset to distinguish between benign and malignant tumors.
- Performed data preprocessing, feature scaling, and feature selection to improve model performance.
- Evaluated Logistic Regression, Support Vector Machine, and Random Forest classifiers using accuracy, precision, recall, and F1-score metrics.
- Achieved classification accuracy above 96% using optimized hyperparameters.

HindsightOps: AI Incident Memory Agent

[GitHub](#)

Next.js, React, TypeScript, FastAPI, OpenAI API, RAG, Vector Embeddings

- Built an AI-powered incident response assistant that combines Large Language Models (LLMs) with organizational memory to help SREs and DevOps teams diagnose incidents faster.
- Designed a Retrieval-Augmented Generation (RAG) pipeline using Vectorize Hindsight for semantic search and similarity-based retrieval of historical incidents from a vector database.
- Implemented automated Root Cause Analysis (RCA) generation, producing incident summaries, timeline reconstruction, root cause identification, impact assessment, and resolution recommendations.

Game Academic Performance Predictor

[GitHub](#)

Python, Scikit-learn, XGBoost, SHAP

- Built an end-to-end machine learning pipeline on an 8,000-student dataset to predict academic grades (regression) and performance bands (classification) from gaming behavior, study habits, and demographic features.
- Engineered domain-specific features, evaluated multiple ML models using 5-fold cross-validation, and achieved an R^2 score of 0.939 with XGBoost.
- Applied SHAP and feature importance analysis to provide interpretable predictions, identifying study-to-gaming balance as the strongest predictor of academic performance.

Repository Evaluation Tool

[GitHub](#)

Python, GitHub API, Data Analysis

- Created a Python-based tool for automated GitHub repository data extraction and quality analysis.
- Generated actionable insights to evaluate repository health, activity, and maintainability for open-source and academic projects.

Education

Bansal Institute of Engineering and Technology

Lucknow, India

B.Tech in Computer Science and Engineering (AI & ML)

2023 – 2027

Career Convent Girls College

Lucknow, India

Intermediate (ISC Board)

2022 – 2023

Career Convent Girls College

Lucknow, India

High School (ICSE Board)

2020 – 2021

Research & Achievements

Google Student Ambassador (2026): Spearheaded technical workshops for 150+ students on AI and cloud platforms.

Research: Authored "CoreSight" paper, presented at the ICSEAIS Conference 2026.

Certifications: GenAI Powered Data Analytics (Forage, 2026); Data Analytics (Deloitte, 2026), ServiceNow Virtual Internship Program (ServiceNow University, 2026) .